
Data Analysis Project

Data Analyst: Marceline Setiawan

Client/Sponsor:

Bellabeat — Urška Sršen, Chief Creative Officer

Purpose:

Bellabeat wants to better understand how consumers use non-Bellabeat smart devices. This project will analyze Fitbit smart-device usage data to identify trends in user activity and movement patterns.

The goal is to use these findings to identify opportunities that could help Bellabeat better understand its customers and inform its marketing strategy. The analysis will also apply the findings to one Bellabeat product and develop a data-driven marketing recommendation.

Scope / Major Project Activities:

Activity	Description
Data Preparation	Collect, organize, and prepare the available Fitbit/Fitabase datasets for analysis.
Data Cleaning	Check for duplicate records, missing values, inconsistent formats, and other data-quality issues.
Data Validation	Validate key fields and compare relevant datasets to ensure the data used for analysis is reliable and consistent.
Exploratory Analysis	Analyze daily steps, activity minutes, activity distance, and other relevant variables to identify patterns and trends.
Data Visualization	Create charts and visualizations to communicate important findings, including activity patterns and differences between participants.
Marketing Recommendations	Apply the findings to a selected Bellabeat product and develop a marketing recommendation based on the observed user behavior.

This project does not include:

- Analysis of Bellabeat's actual sales or revenue data.
- Direct surveys, interviews, or primary research with Bellabeat customers.
- Predicting future sales or guaranteeing the success of a marketing campaign.

Deliverables:

Deliverable	Description/ Details
Cleaned Dataset	A cleaned and prepared Fitbit/Fitabase dataset used for analysis.
SQL Analysis	SQL queries documenting data cleaning, validation, and selected analysis.
Data Visualizations	Charts showing key trends and relationships identified in the data.
Analysis Findings	A summary of the main activity patterns and differences observed among participants.
Marketing Recommendation	A recommendation for applying the findings to one Bellabeat product and informing its marketing strategy.
Final Presentation	A stakeholder presentation summarizing the business question, methodology, findings, implications, and recommendations.

Schedule Overview / Major Milestones:

Milestone	Expected Completion Date	Description/Details
Define Business Task	19 September 2026	Define the business problem, stakeholders, objectives, and key questions.
Data Preparation	21 September 2026	Collect, organize, and prepare the Fitbit/Fitabase datasets.
Data Cleaning & Validation	22 September 2026	Identify and address data-quality issues and validate the analysis dataset.
Exploratory Analysis	25 September 2026	Analyze activity levels, steps, distance, and participant

		variation.
Visualization	27 September 2026	Create visualizations to communicate key findings and relationships.
Develop Recommendations	28 September 2026	Apply findings to a Bellabeat product and develop a marketing recommendation.
Prepare Final Presentation	29 September 2026	Organize findings, recommendations, and supporting visualizations into a stakeholder presentation.
Final Review	30 September 2026	Review the analysis, presentation, and supporting documentation for accuracy and completeness.

*Estimated date for completion:

September 30, 2026